

Building Atoms: Structure, Symbols and Isotopes

PhET Build an Atom · PhET

Years 9-10

~30 minutes

Science · Chemistry

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/build-an-atom>

Curriculum links: AC9S9U06 · AC9S10U06 · atomic structure

Learning intentions

- I can build an atom of a given element by adding the correct number of protons, neutrons and electrons.
- I can explain how the number of protons, neutrons and electrons determines an atom's element, mass number and charge.
- I can use the periodic table to check whether my constructed atom is correct.

Getting started

Open PhET Build an Atom and go to the Atom screen. Start with an empty atom, with no particles added, and check the element name shown, which should read blank or 'no element'. Keep the mass number, charge and element symbol displays turned on as you work.

Tasks

1. Add protons one at a time to your empty atom and watch the element name change. How many protons does it take to make carbon?

2. With 6 protons in place, add 6 neutrons and 6 electrons. Record the resulting mass number and overall charge.

3. Remove one electron from your carbon atom, keeping protons and neutrons the same. What happens to the charge, and what is this particle now called?

4. Build four different atoms or isotopes of your choice and record the particles you used and the resulting mass number.

Element	Protons	Neutrons added	Resulting mass number

5. Two atoms both have 6 protons, but one has 6 neutrons and the other has 8 neutrons. Explain whether these are the same element, and what term describes their relationship.

6. Switch to the Symbol screen. Build an atom to match a given isotope notation, for example an atom with mass number 14 and atomic number 6. Record the protons, neutrons and electrons you used.

7. Explain, using what you built in this simulation, how the arrangement of the periodic table relates to the number of protons in an atom.

Extension (optional)

Build an ion with a 2+ charge from an element of your choice. Explain what you changed to create the charge, and why removing electrons, rather than protons, is what forms an ion.